화학수학

열여덟 번째 수업

확률

Probability Distribution (확률분포)

mathematical function that gives

the probabilities of occurrence of

different possible outcomes for an experiment

- Discrete probability distribution (이산 확률분포): possible outcomes have discrete values
- Continuous probability distribution (연속 확률분포): possible outcomes have continuous values

Not in Mortimer

Probability Distributions

	Discrete	Continuous
Variable	$x_1, x_2, \cdots, x_n$	x: real in $[x_{\min}, x_{\max}]$
Probability distribution	p_i	$f(x)\,\mathrm{d}x$
Normalization	$\sum_{i=1}^{n} p_i = 1$	$\int_{x_{\min}}^{x_{\max}} f(x)\,\mathrm{d}x = 1$
Mean value (normalized dist.)	$\langle x \rangle = \sum_{i=1}^{n} x_i p_i$	$\langle x \rangle = \int_{x_{\min}}^{x_{\max}} x f(x)\,\mathrm{d}x$
Expected value (normalized dist.)	$\langle g \rangle = \sum_{i=1}^{n} g(x_i) p_i$	$\langle g \rangle = \int_{x_{\min}}^{x_{\max}} g(x) f(x)\,\mathrm{d}x$

Not in Mortimer

Examples

- Discrete probability distribution
  - Binomial probability distribution
- Continuous probability distribution
  - Uniform probability distribution
  - Gaussian probability distribution

Binomial Probability Distribution

- p = probability that "heads" will occur
- $(1-p)$ = probability that "tails" will occur

$$P(n,m) = \frac{n!}{m!\,(n-m)!}p^m(1-p)^{n-m}$$

Not in Mortimer

Mean Value and Variance

- Mean value

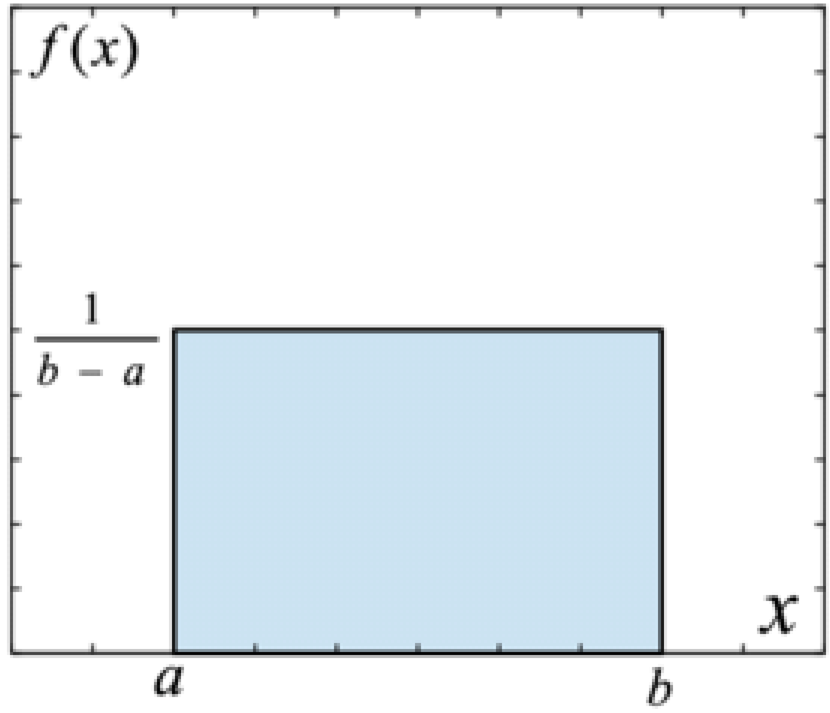
$$\langle x \rangle = \sum_{i=1}^{\infty} p_i x_i = \sum_{m=0}^{n} \frac{n!}{m!\,(n-m)!} p^m (1-p)^{n-m} \times m$$

$$\langle x \rangle = np$$

- Variance

$$\sigma_x^2 = \langle x^2 \rangle - \langle x \rangle^2 = np(1-p)$$

Uniform Probability Distribution



(Image: https://www.researchgate.net/figure/Probability-distribution-function-a-uniform-distribution-of-the-excitation-frequency_fig5_339429775)

Mean Value and Variance

- Mean value

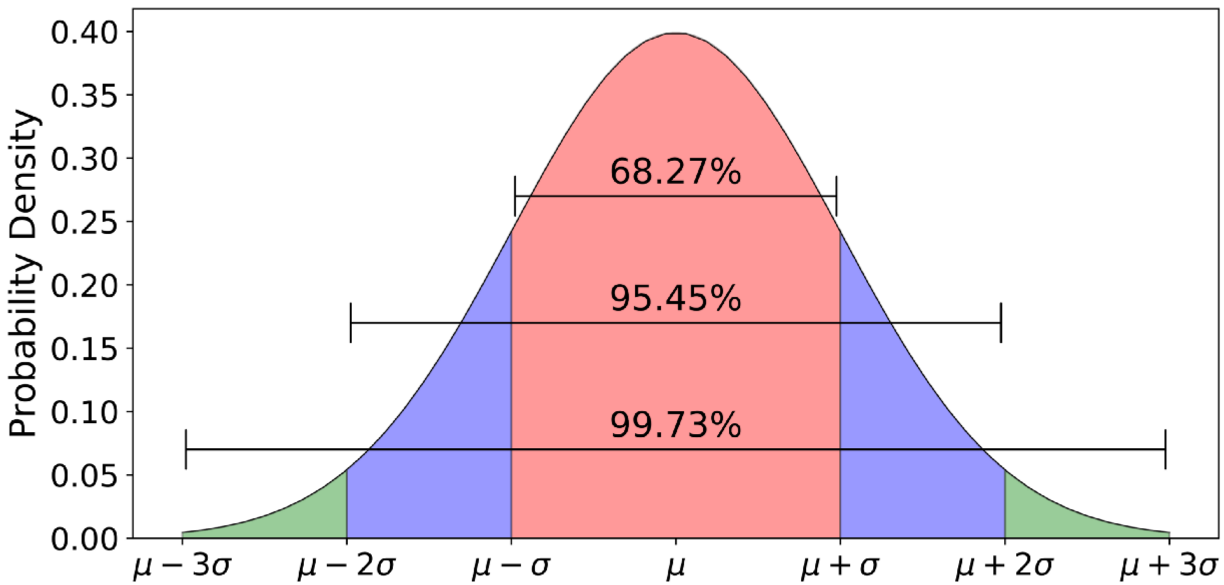
$$\langle x \rangle = \frac{a+b}{2}$$

- Variance

$$\sigma_x^2 = \frac{(a-b)^2}{12}$$

Gaussian Probability Distribution

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]$$



(Image: https://towardsdatascience.com/understanding-the-68-95-99-7-rule-for-a-normal-distribution-b7b7cbf760c2)

Mean Value and Variance

- Mean value

$$\langle x \rangle = \int_{-\infty}^{\infty} x f(x)\, \mathrm{d}x = \mu$$

- Variance

$$\sigma_x^2 = \langle x^2 \rangle - \langle x \rangle^2 = \sigma^2$$

Central Limit Theorem (중심극한정리)

For random variables $x_1, x_2, \cdots, x_n$,

$$y = x_1 + x_2 + \cdots + x_n$$

is governed by a probability distribution that approaches a Gaussian distribution for large n

- If experimental errors arise from multiples sources, they should be at least approximately described by a Gaussian distribution

오늘의 수업

- Applications of probability theory
  - Quantum mechanics
  - Kinetic theory of a gas

Application: Quantum Mechanics

For a wave function ψ of the system,

$$\text{Probability distribution} \propto |\psi|^2$$

e.g. Ground state of a 1-dimensional particle in a box

- Wave function $\psi(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right)$
- Probability distribution $f(x) = |\psi|^2 = \frac{2}{L} \sin^2\left(\frac{\pi x}{L}\right)$

Modified Example 15.9

For a particle in a box of length L in its ground state, find the probability that the particle will be found in $(0, L/2)$.

$$\text{Probability} = \int_0^{L/2} f(x)\,\mathrm{d}x = \int_0^{L/2} |\psi(x)|^2\,\mathrm{d}x$$

Since $f(x) = |\psi|^2 = \frac{2}{L}\sin^2\left(\frac{\pi x}{L}\right)$,

$$\text{Probability} = \int_0^{L/2} \frac{2}{L}\sin^2\left(\frac{\pi x}{L}\right)\mathrm{d}x = \frac{2}{\pi}\int_0^{\pi/2} \sin^2 t\,\mathrm{d}t = \frac{1}{2}$$

Not in Mortimer

Operators in Quantum Mechanics

"For every observable quantity,

there is a corresponding mathematical operator"

- Position operators: $\hat{x} = x,\ \hat{y} = y,\ \hat{z} = z$
- Momentum operators:

$$\hat{p}_x = \frac{\hbar}{i}\frac{\partial}{\partial x}, \qquad \hat{p}_y = \frac{\hbar}{i}\frac{\partial}{\partial y}, \qquad \hat{p}_z = \frac{\hbar}{i}\frac{\partial}{\partial z}$$

- System energy operator (Hamiltonian):

$$\hat{H} = -\frac{\hbar}{2m}\nabla^2 + \mathcal{V}(x, y, z)$$

Not in Mortimer

Quantum Measurement

If a wavefunction ψ of the system satisfies

$$\hat{A}\psi = a\psi$$

for a certain operator $\hat{A}$,

the measurement of the corresponding observable always gives a (eigenvalue).

Not in Mortimer

If Wavefunction ≠ Eigenfunction?

If a wavefunction is represented as linear combination of the eigenfunctions of an operator ($\hat{A}f_k = a_k f_k$),

$$\psi = c_1 f_1 + c_2 f_2 + c_3 f_3 + \cdots + c_n f_n,$$

the measurement gives the eigenvalue a_i with the probability of the squared coefficient $|c_i|^2$.

Expectation Value (기댓값)

"The predicted mean value of

many measurements of an observable"

(expected value)

For $\psi = c_1 f_1 + c_2 f_2 + c_3 f_3 + \cdots + c_n f_n$,

$$\begin{aligned} \langle A \rangle &= a_1 P_1 + a_2 P_2 + \cdots + a_n P_n \\ &= a_1 |c_1|^2 + a_2 |c_2|^2 + \cdots + a_n |c_n|^2 \end{aligned}$$

Expectation Value (기댓값)

For a wave function ψ of the system,

$$\langle A \rangle = \int \psi^* \hat{A} \psi \, \mathrm{d}q$$

where $\hat{A}$ is the operator corresponding to the variable A.

Example 15.8

Find the expressions for $\langle x \rangle$ and σ_x for a particle in a box of length L in its ground state.

$$\psi(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right)$$

The expectation values are

$$\langle x \rangle = \int_0^L \psi^*(x)\hat{x}\psi(x)\,\mathrm{d}x = \int_0^L x|\psi|^2\,\mathrm{d}x = \frac{2}{L}\int_0^L x\sin^2\left(\frac{\pi x}{L}\right)\mathrm{d}x = \frac{L}{2}$$

$$\langle x^2 \rangle = \int_0^L \psi^*(x)\hat{x}^2\psi(x)\,\mathrm{d}x = \int_0^L x^2|\psi|^2\,\mathrm{d}x = \frac{2}{L}\int_0^L x^2\sin^2\left(\frac{\pi x}{L}\right)\mathrm{d}x = \frac{L^2}{3} - \frac{L^2}{2\pi^2}$$

$$\sigma_x = \sqrt{\sigma_x^2} = \sqrt{\langle x^2 \rangle - \langle x \rangle^2} = \sqrt{\frac{L^2}{3} - \frac{L^2}{2\pi^2} - \frac{L^2}{4}} = L\sqrt{\frac{1}{12} - \frac{1}{2\pi^2}}$$

1-Dimensional Hamiltonian Operator

$$\hat{H} = -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + \mathcal{V}(x)$$

- Corresponds to the system energy
- For a particle in a box, $\mathcal{V}(x) = 0$

Example 15.10

Find the expectation value of the energy and its standard deviation for the ground state of our particle in a box.

$$\psi(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{\pi x}{L}\right); \qquad \hat{H} = -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2}$$

$$\langle E\rangle = \int_0^L \psi^*(x)\hat{H}\psi(x)\,\mathrm{d}x = \int_0^L \sqrt{\frac{2}{L}}\sin\left(\frac{\pi x}{L}\right)\left[-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2}\left\{\sqrt{\frac{2}{L}}\sin\left(\frac{\pi x}{L}\right)\right\}\right]\mathrm{d}x$$

$$= -\frac{\hbar^2}{mL}\int_0^L \sin\left(\frac{\pi x}{L}\right)\left(-\frac{\pi^2}{L^2}\right)\sin\left(\frac{\pi x}{L}\right)\mathrm{d}x = \frac{\pi^2\hbar^2}{mL^3}\int_0^L \sin^2\left(\frac{\pi x}{L}\right)\mathrm{d}x$$

$$= \frac{\pi^2\hbar^2}{mL^3}\times\frac{L}{2} = \frac{\pi^2\hbar^2}{2mL^2} = \frac{h^2}{8mL^2}$$

Example 15.10

Find the expectation value of the energy and its standard deviation for the ground state of our particle in a box.

$$\psi(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{\pi x}{L}\right); \qquad \hat{H} = -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2}$$

$$\langle E^2\rangle = \int_0^L \psi^*(x)\hat{H}^2\psi(x)\,\mathrm{d}x = \int_0^L \sqrt{\frac{2}{L}}\sin\left(\frac{\pi x}{L}\right)\left[\left(\frac{\hbar^2}{2m}\right)^2\frac{\mathrm{d}^4}{\mathrm{d}x^4}\left\{\sqrt{\frac{2}{L}}\sin\left(\frac{\pi x}{L}\right)\right\}\right]\mathrm{d}x$$

$$= \frac{\hbar^4}{2m^2L}\int_0^L \sin\left(\frac{\pi x}{L}\right)\left(\frac{\pi^4}{L^4}\right)\sin\left(\frac{\pi x}{L}\right)\mathrm{d}x = \frac{\pi^4\hbar^4}{2m^2L^5}\int_0^L \sin^2\left(\frac{\pi x}{L}\right)\mathrm{d}x$$

$$= \frac{\pi^4\hbar^4}{2m^2L^5}\times\frac{L}{2} = \frac{\pi^4\hbar^4}{4m^2L^4} = \frac{h^4}{64m^2L^4}$$

Example 15.10

Find the expectation value of the energy and its standard deviation for the ground state of our particle in a box.

$$\langle E \rangle = \frac{h^2}{8mL^2}; \qquad \langle E^2 \rangle = \frac{h^4}{64m^2L^4}$$

The standard deviation is

$$\sigma_E = \sqrt{\sigma_E^2} = \sqrt{\langle E^2 \rangle - \langle E \rangle^2} = \sqrt{\frac{h^4}{64m^2L^4} - \left(\frac{h^2}{8mL^2}\right)^2} = 0$$

Application: Kinetic Theory of a Gas

- Boltzmann factor: in the thermal equilibrium of temperature T, the probability of having energy E is

$$f(E) \propto e^{-E/k_{\mathrm{B}}T}$$

- The probability that a gas molecule has the velocity (v_x, v_y, v_z) is

$$f(v_x, v_y, v_z) \propto \exp\left(-\frac{m\left(v_x^2 + v_y^2 + v_z^2\right)}{2k_{\mathrm{B}}T}\right)$$

Not in Mortimer

Normalization

$$f(v_x, v_y, v_z) = A \exp\left(-\frac{m\left(v_x^2 + v_y^2 + v_z^2\right)}{2k_\mathrm{B}T}\right)$$

$$I = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(v_x, v_y, v_z)\,\mathrm{d}v_x\mathrm{d}v_y\mathrm{d}v_z = 1$$

$$I = A\int_{-\infty}^{\infty} e^{-mv_x^2/2k_\mathrm{B}T}\,\mathrm{d}v_x \int_{-\infty}^{\infty} e^{-mv_y^2/2k_\mathrm{B}T}\,\mathrm{d}v_y \int_{-\infty}^{\infty} e^{-mv_z^2/2k_\mathrm{B}T}\,\mathrm{d}v_z$$

$$I = A\left(\sqrt{\frac{2\pi k_\mathrm{B}T}{m}}\right)^3 \to A = \left(\frac{m}{2\pi k_\mathrm{B}T}\right)^{3/2}$$

Velocity to Speed

- $f(v_x, v_y, v_z)\, \mathrm{d}v_x \mathrm{d}v_y \mathrm{d}v_z$:

  Probability that a molecule has a velocity in the range

  $$[v_x, v_x + \mathrm{d}v_x], [v_y, v_y + \mathrm{d}v_y], [v_z, v_z + \mathrm{d}v_z]$$

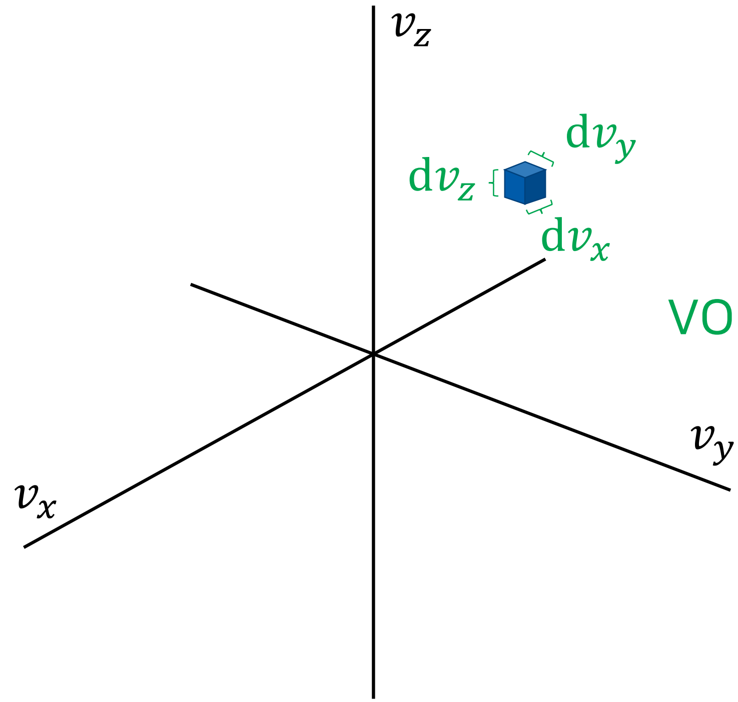
- $f_v(v)\, \mathrm{d}v$:

  Probability that a molecule has a speed in the range

  $$[v, v + \mathrm{d}v]$$

Can we obtain $f_v(v)\mathrm{d}v$ from $f(v_x, v_y, v_z)\mathrm{d}v_x \mathrm{d}v_y \mathrm{d}v_z$?

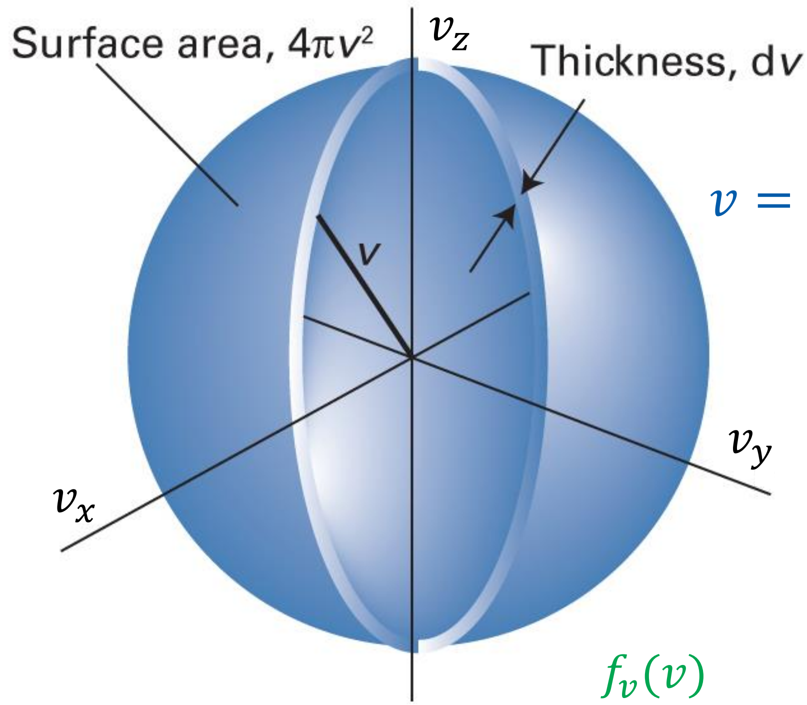
Velocity Space



volume: $\mathrm{d}v_x\mathrm{d}v_y\mathrm{d}v_z$

$f(v_x, v_y, v_z)\,\mathrm{d}v_x\mathrm{d}v_y\mathrm{d}v_z$ = function × volume

Velocity Space



$$v = \sqrt{v_x^2 + v_y^2 + v_z^2}$$

$$\text{function} \times \text{volume} = \overbrace{f\left(v_x, v_y, v_z\right) \times 4\pi v^2 \mathrm{d}v}^{f_v(v)}$$

(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 11th ed., Figure 1B.3)

Maxwell-Boltzmann Distribution

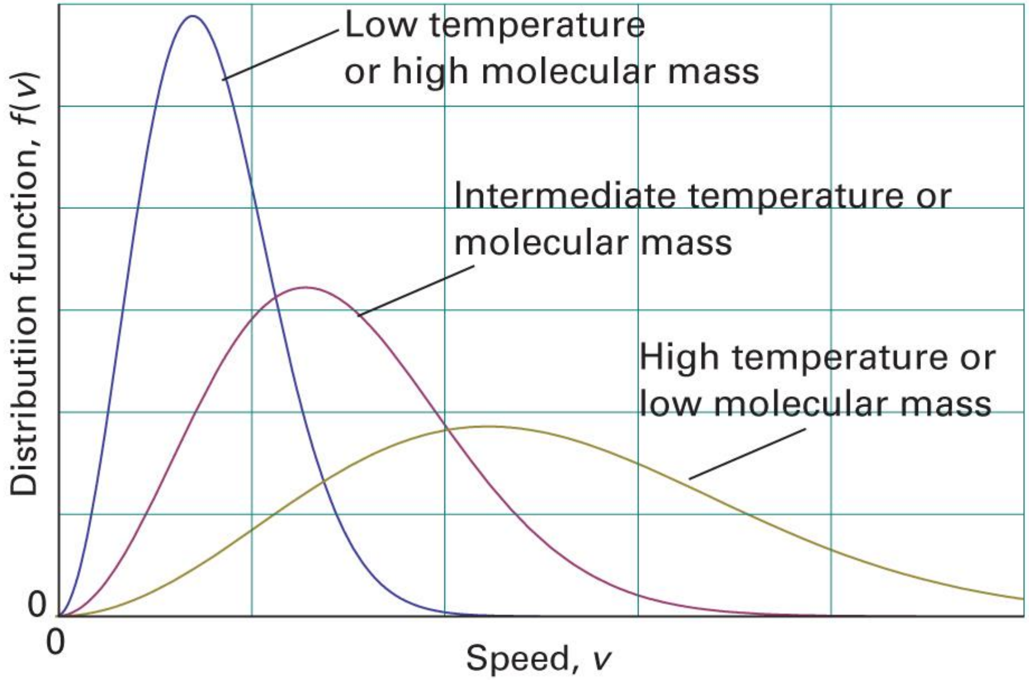
$$f_v(v) = f\left(v_x, v_y, v_z\right)4\pi v^2$$

$$f_v(v) = 4\pi\left(\frac{m}{2\pi k_\mathrm{B}T}\right)^{3/2} v^2 e^{-m\left(v_x^2+v_y^2+v_z^2\right)/2k_\mathrm{B}T}$$

$$f_v(v) = 4\pi\left(\frac{m}{2\pi k_\mathrm{B}T}\right)^{3/2} v^2 e^{-mv^2/2k_\mathrm{B}T}$$

Behaviors of the Distribution

$$f_v(v) = 4\pi\left(\frac{m}{2\pi k_\mathrm{B}T}\right)^{3/2} v^2 e^{-mv^2/2k_\mathrm{B}T}$$



(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 11th ed., Figure 1B.4)

Mean Speed (평균 속력)

$$f_v(v) = 4\pi\left(\frac{m}{2\pi k_{\mathrm{B}}T}\right)^{3/2} v^2 e^{-mv^2/2k_{\mathrm{B}}T}$$

$$\langle v\rangle = \int_0^\infty v f_v(v)\,\mathrm{d}v = \int_0^\infty 4\pi\left(\frac{m}{2\pi k_{\mathrm{B}}T}\right)^{3/2} v^3 e^{-mv^2/2k_{\mathrm{B}}T}\,\mathrm{d}v$$

$$= 4\pi\left(\frac{m}{2\pi k_{\mathrm{B}}T}\right)^{3/2} \int_0^\infty v^3 e^{-mv^2/2k_{\mathrm{B}}T}\,\mathrm{d}v$$

$$= 4\pi\left(\frac{m}{2\pi k_{\mathrm{B}}T}\right)^{3/2} \frac{(2k_{\mathrm{B}}T)^2}{2m^2} = \left(\frac{8k_BT}{\pi m}\right)^{1/2}$$

Example 15.12

Find a formula for the mean of the square of the speed and the root-mean-square speed of molecules in a gas.

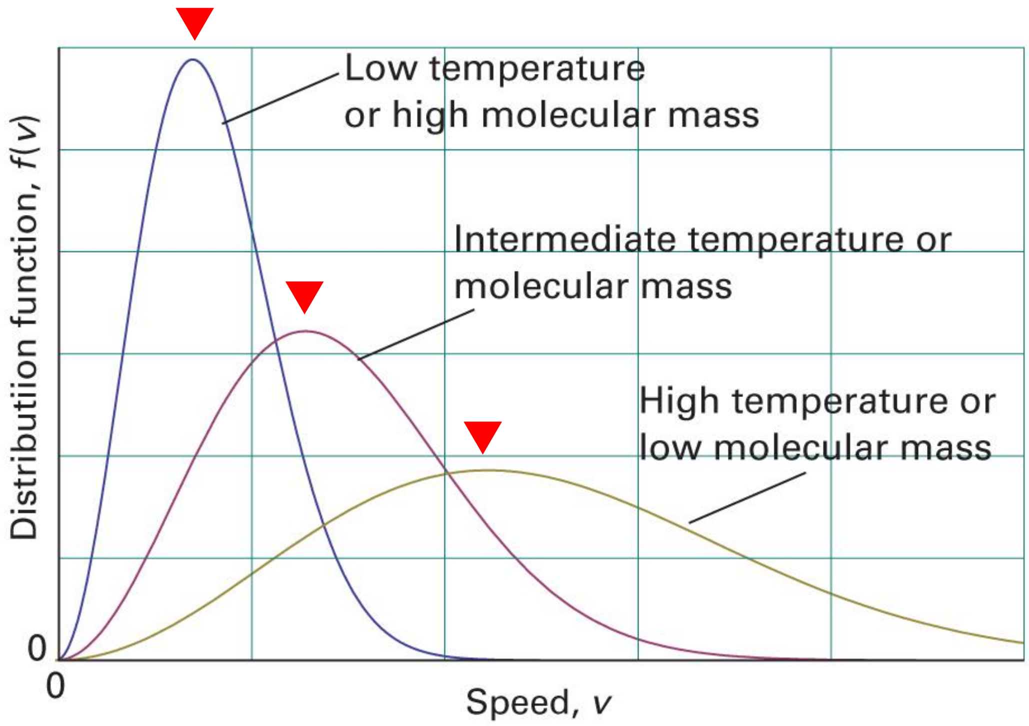
$$f_v(v) = 4\pi \left(\frac{m}{2\pi k_\mathrm{B} T}\right)^{3/2} v^2 e^{-mv^2/2k_\mathrm{B}T}$$

$$\langle v^2 \rangle = \int_0^\infty v^2 f_v(v)\,\mathrm{d}v = \int_0^\infty 4\pi \left(\frac{m}{2\pi k_\mathrm{B} T}\right)^{3/2} v^4 e^{-mv^2/2k_\mathrm{B}T}\,\mathrm{d}v$$

$$= 4\pi \left(\frac{m}{2\pi k_\mathrm{B} T}\right)^{3/2} \int_0^\infty v^4 e^{-mv^2/2k_\mathrm{B}T}\,\mathrm{d}v = \frac{3k_\mathrm{B}T}{m}$$

$$v_\mathrm{rms} = \sqrt{\langle v^2 \rangle} = \left(\frac{3k_\mathrm{B}T}{m}\right)^{1/2}$$

Most Probable Speed (최빈 속력)



$$\frac{\mathrm{d}f(v)}{\mathrm{d}v} = 0$$

(Image: Atkins *et al.*, *Atkins' Physical Chemistry*, 11th ed., Figure 1B.4)

Most Probable Speed (최빈 속력)

$$f_v(v) = 4\pi\left(\frac{m}{2\pi k_\mathrm{B}T}\right)^{3/2} v^2 e^{-mv^2/2k_\mathrm{B}T}$$

$$\frac{\mathrm{d}f_v(v)}{\mathrm{d}v} = 4\pi\left(\frac{m}{2\pi k_B T}\right)^{3/2}\left(2ve^{-mv^2/2k_\mathrm{B}T} - \frac{mv^3}{k_\mathrm{B}T}e^{-mv^2/2k_\mathrm{B}T}\right) = 0$$

$$2v - \frac{mv^3}{k_\mathrm{B}T} = 0$$

$$v_\mathrm{mp} = \left(\frac{2k_BT}{m}\right)^{1/2}$$